Pranshu Gupta

https://pranshug.com | linkedin.com/in/pranshug258

Email: pranshug258@outlook.com
Mobile: +1 (425) 679-2402

Professional Experience

- **Microsoft AI** / *Member of Technical Staff / Redmond, WA* — *Aug 2025 – Present*
  - Reduced production GPU footprint by **1,320 H100 GPUs** by analyzing the inference pipeline and eliminating non-essential classifiers and summarizers on the GPT-5.1 instant path.
  - Led end-to-end rollout of **GPT-5.1 instant and reasoning models** to production by designing prompt templates, constrained-sampling, chain of thought handling, and safety integrations across six partner teams; owned launch-readiness evaluations with the Response Quality squad, including **30% improvement in personality evals**.
  - Designed and shipped **Adaptive Thinking**, a dynamic system that adjusts the thinking effort per message based on intent/capability classification to control reasoning token usage.
  - Architected a request-scoped **feature flag system** for the inference API, adopted by MAI and M365 as the standard mechanism for A/B experiments and incremental rollouts.
  - Improved session-management reliability by fixing affinity-drain ordering, eliminating deploy-time connection failures, and reducing **Redis write volume by 9%** via session-affinity deduplication.
  - Built **Grafana dashboards** for LLM API traffic classification across MAI and M365 Copilot, instrumenting dimensions including product, client app, agent, inference intent, license tier, and traffic type; implemented production failure-rate monitor with alerting for Bing Search API dependency.
- **Microsoft Azure AI** / *Senior Software Engineer / Redmond, WA* — *Jan 2025 – Aug 2025*
  - Designed a distributed **token-based rate limiter** using Tiktoken for precise token counting, replacing a length-based estimator with **219% average error**; implemented pub-sub quota state synchronization across instances.
  - Contributed to an intelligent **model router** for per-request model selection based on cost/complexity signals, reducing per-inference cost by **55%**.
  - Designed **multi-region routing architecture** for the M365 LLM inference API: same-region-preferred routing with ordered fallback regions to balance GPU session affinity against compute availability spillover across global deployments.
- **Microsoft Azure Core** / *Senior Software Engineer / Redmond, WA* — *Mar 2024 – Jan 2025*
  - Designed and shipped a token-based job throttling system with multi-bucket quota enforcement and retry-storm protection, preventing partition-level noisy-neighbor throttling in background-job infrastructure of Azure Control Plane.
  - Single-handedly designed and shipped async operation URI signing, adopted by **242 Azure service teams** and enforced for **220+ resource providers**.
- **Microsoft Azure Core** / *Software Engineer II / Redmond, WA* — *Jun 2020 – Mar 2024*
  - Designed and shipped async operation callbacks replacing polling, reducing ARM-to-resource-provider traffic by up to **94%** (130M fewer requests/day) and improving p95 latency by **12 seconds** for Azure Container Instances.
  - Extended and maintained **Azure DevOps CI/CD pipelines** for ARM control-plane services with staged multi-region rollouts, 24-hour regional bake times, functional test gates, and rollback-via-redeploy across global Azure regions.
  - Led **Azure region teardown** end-to-end: built workflows to halt new resource creation, resolve subscription/resource-group hierarchies, and migrate live resources to healthy regions with zero data loss.
- **Microsoft Core Services** / *Software Engineer / Hyderabad, India* — *Jun 2017 – Aug 2019*
  - Migrated customer data enrichment service to serverless design, reducing cost by **90%** and improving throughput by **3x**.

Technical Skills

- **AI/ML & Inference**: LLM inference/serving, prompt engineering, model evaluations, safety orchestration, RAG
- **Languages & Systems**: Python, C#, C++, KQL, OpenAI API, Kubernetes, Redis, Cosmos DB, Service Bus, .NET
- **Cloud & Infrastructure**: Azure (ARM, AKS, Service Bus, Cosmos DB), AWS (Bedrock, S3, Textract, ECR), ARM Templates, Azure DevOps, Grafana

Selected Projects

- **AI Lab Assistant** (TurnTheBus nonprofit, 2025–Present): Architected and built end-to-end on **AWS** (Bedrock/Claude Haiku, S3, Textract, ECR); uses **RAG** over physics lab manuals to generate Socratic hints that guide high school students through virtual experiments based on live experiment state.

Technical Writing

- *Runtime Optimizations for LLMs* (2026). Production notes on model offloading, KV cache management, and quantization for latency/cost-efficient LLM serving. pranshug.com/blog/runtime-optimizations-for-llms

Education

- **Georgia Institute of Technology** / *M.S. Computer Science (Machine Learning) / Atlanta, GA* — *May 2020*
- **Indian Institute of Technology, Kanpur** / *B.Tech, C.S. and Eng. / Kanpur, India* — *May 2017*